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SNOMED CT EXPO 2026 · LIGHTNING TALK

Data Golf 2026

A community analytics challenge
- learnings and insights

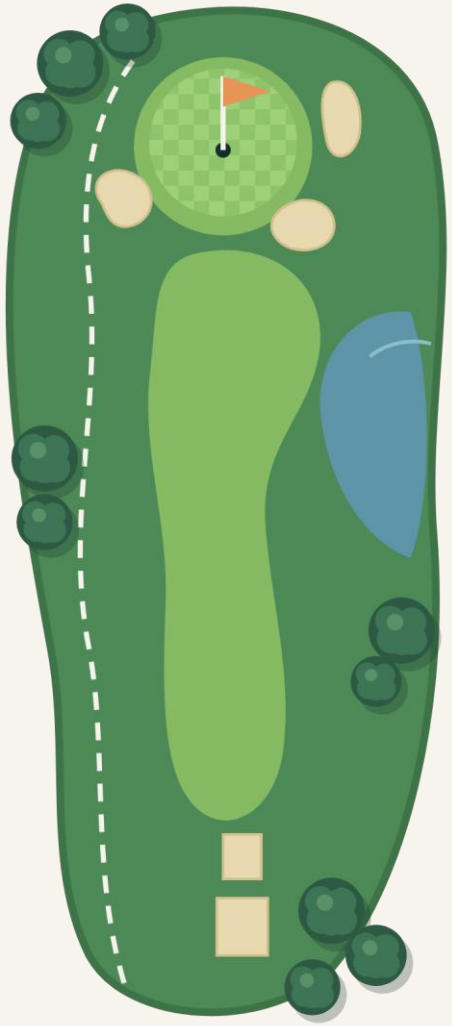


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The Competition

279,912	synthetic patient encounters, 2021 to 2026
6	SNOMED CT-AU editions; each encounter answered with the edition of its date
18	yes/no questions, scored by macro F1
13	complete submissions (15 received)
0.0024	macro F1 between the top three (0.9922, 0.9914, 0.9898); ten of 13 scored above 0.91

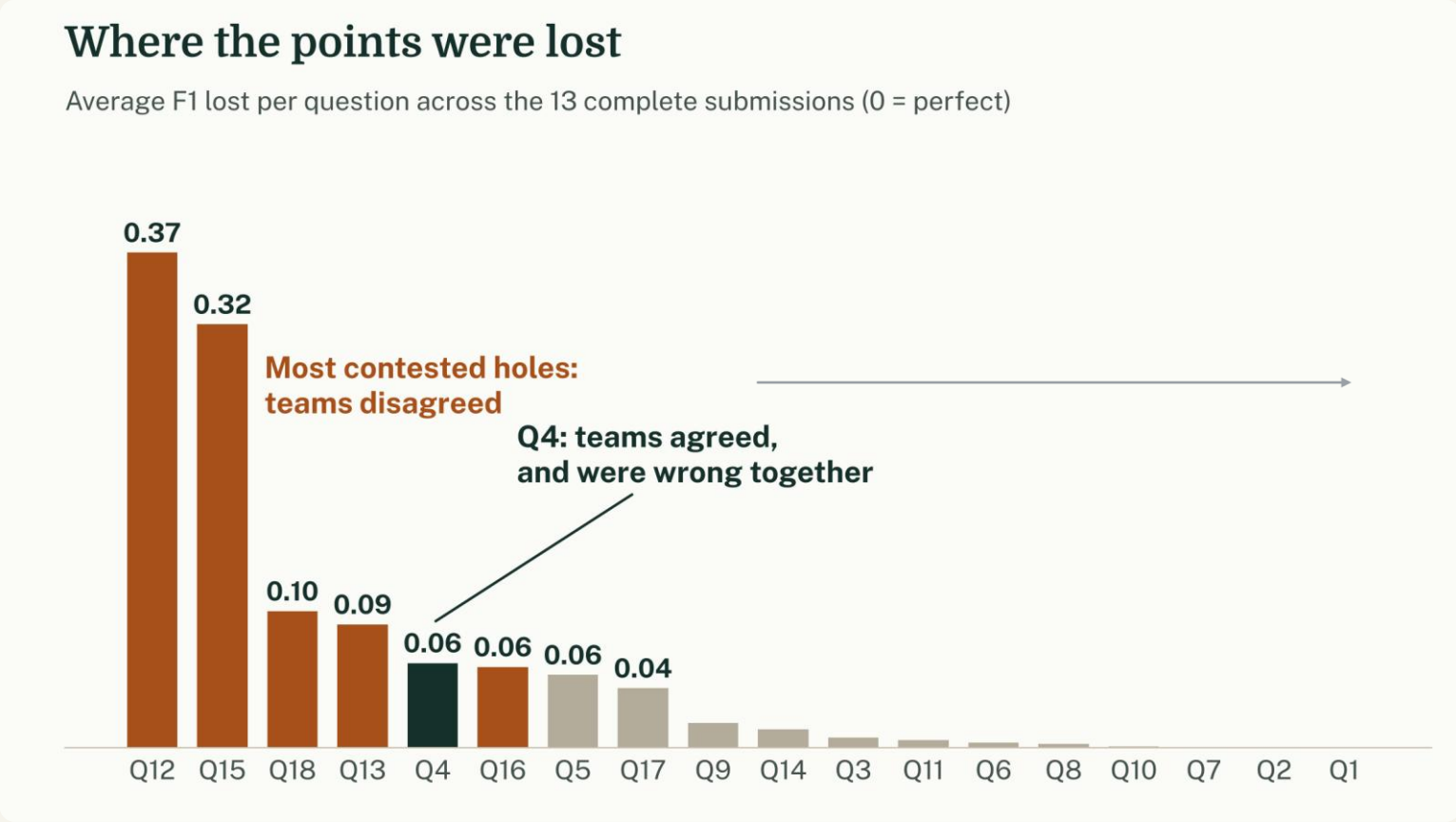


The course: 18 “holes”

Q1	Female as recorded sex	
Q2	BMI 30+ and cholesterol:HDL above 5	
Q3	Healthy or no current problems	
Q4	Hyperthyroidism per clinical note	
Q5	Osteoarthritis in 2024	
Q6	Taking an ACE inhibitor	
Q7	Taking a cardiac glycoside	
Q8	Fungal foot infection	
Q9	Two or more paracetamol products	
Q10	Bilateral condition	
Q11	Female 15 to 49 on methotrexate	
Q12	**Pregnant and on a teratogen**	**#1**
Q13	**OA or RA, oral NSAID, eGFR below 60**	**#5**
Q14	Upper limb wound, not fingers/thumb	
Q15	**Pregnant, fasting glucose 5.1+ mmol/L**	**#2**
Q16	**Non-Strep bacterial joint infection**	**#4**
Q17	Taking an oral iron supplement	
Q18	**Mineral supplement, not Na, K or Cl**	**#3**

Dark rows: the five holes where teams disagreed most, ranked #1 to #5.

Most holes were par. Five decided the round.



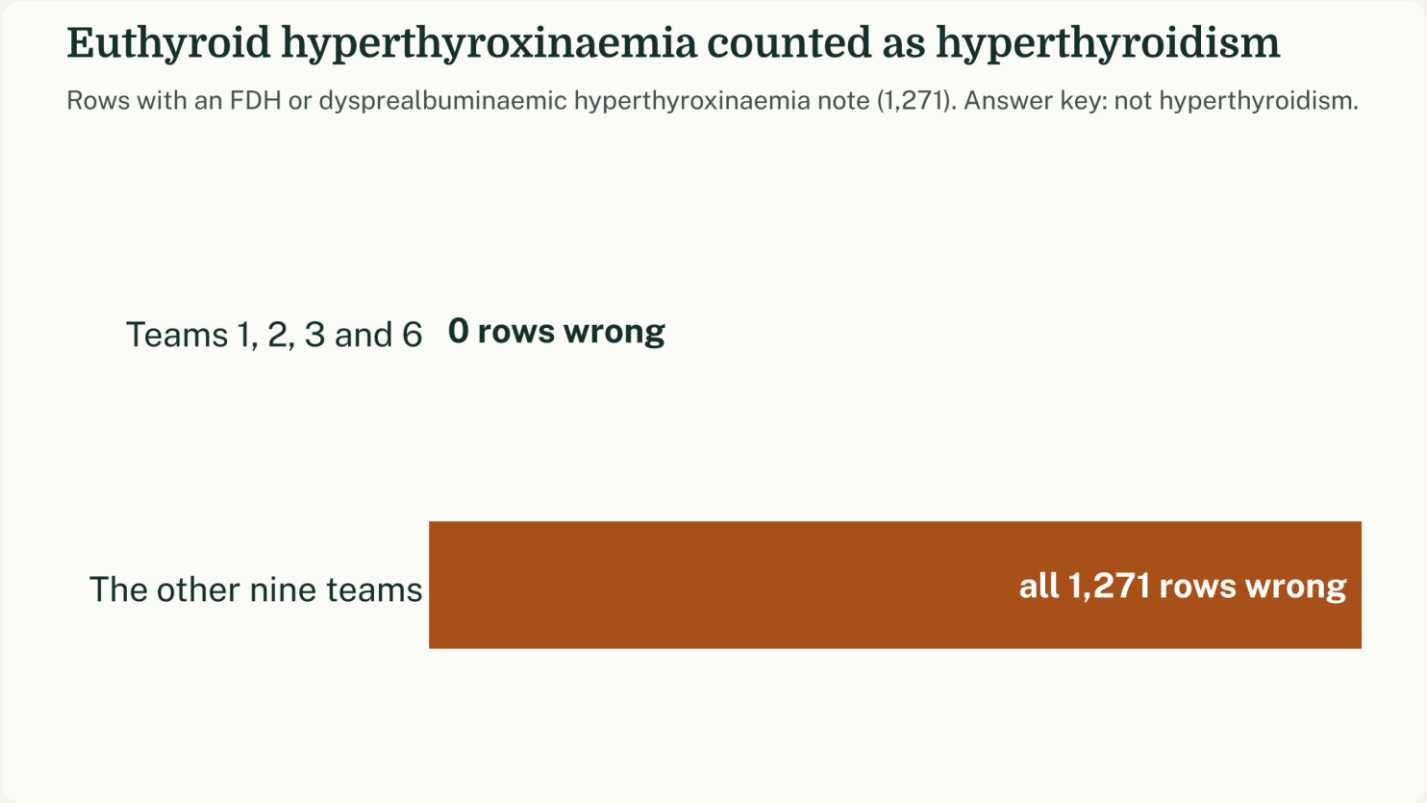
Q1, Q2 and Q7: **every team perfect**.

Losses sit in the **two pregnancy questions**, mineral supplements, joint infection and arthritis with NSAIDs.

But the biggest **shared** mistake, on Q4, was not contested at all.

Q4 · HYPERTHYROIDISM

The hierarchy said yes. The clinician says no.



9 of 13 teams counted familial dysalbuminaemic hyperthyroxinaemia as hyperthyroidism.

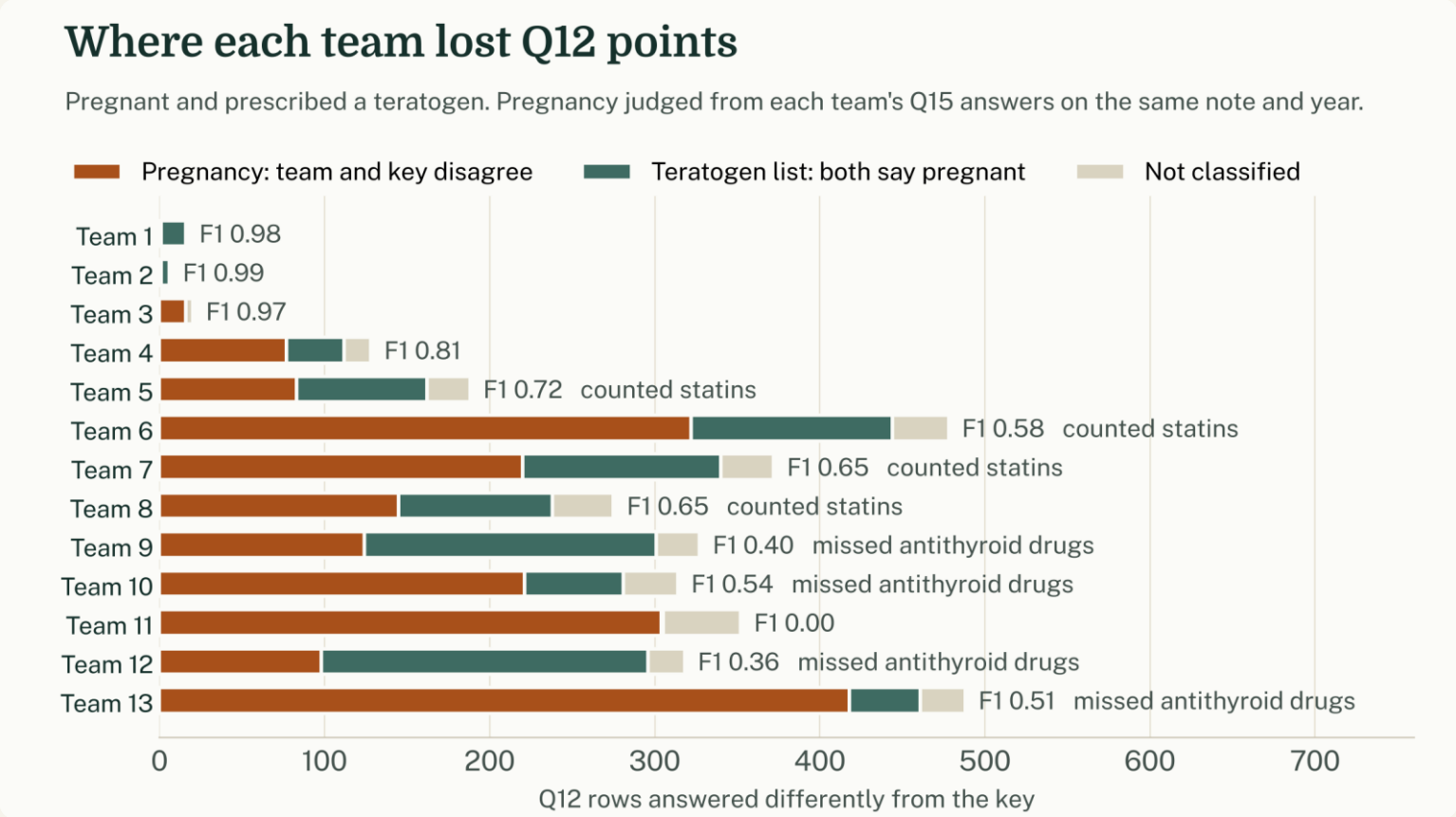
It sits under *Hyperthyroidism*, but patients are euthyroid **[1,2]**.

Seven teams gave an **identical** answer column.

[1] Pappa T, et al. Best Pract Res Clin Endocrinol Metab 2015;29:735-47. doi:10.1016/j.beem.2015.09.002
[2] Kragh-Hansen U, et al. Front Endocrinol 2017;8:297. doi:10.3389/fendo.2017.00297

Q12 · PREGNANT AND PRESCRIBED A TERATOGEN

No ECL returns a teratogen



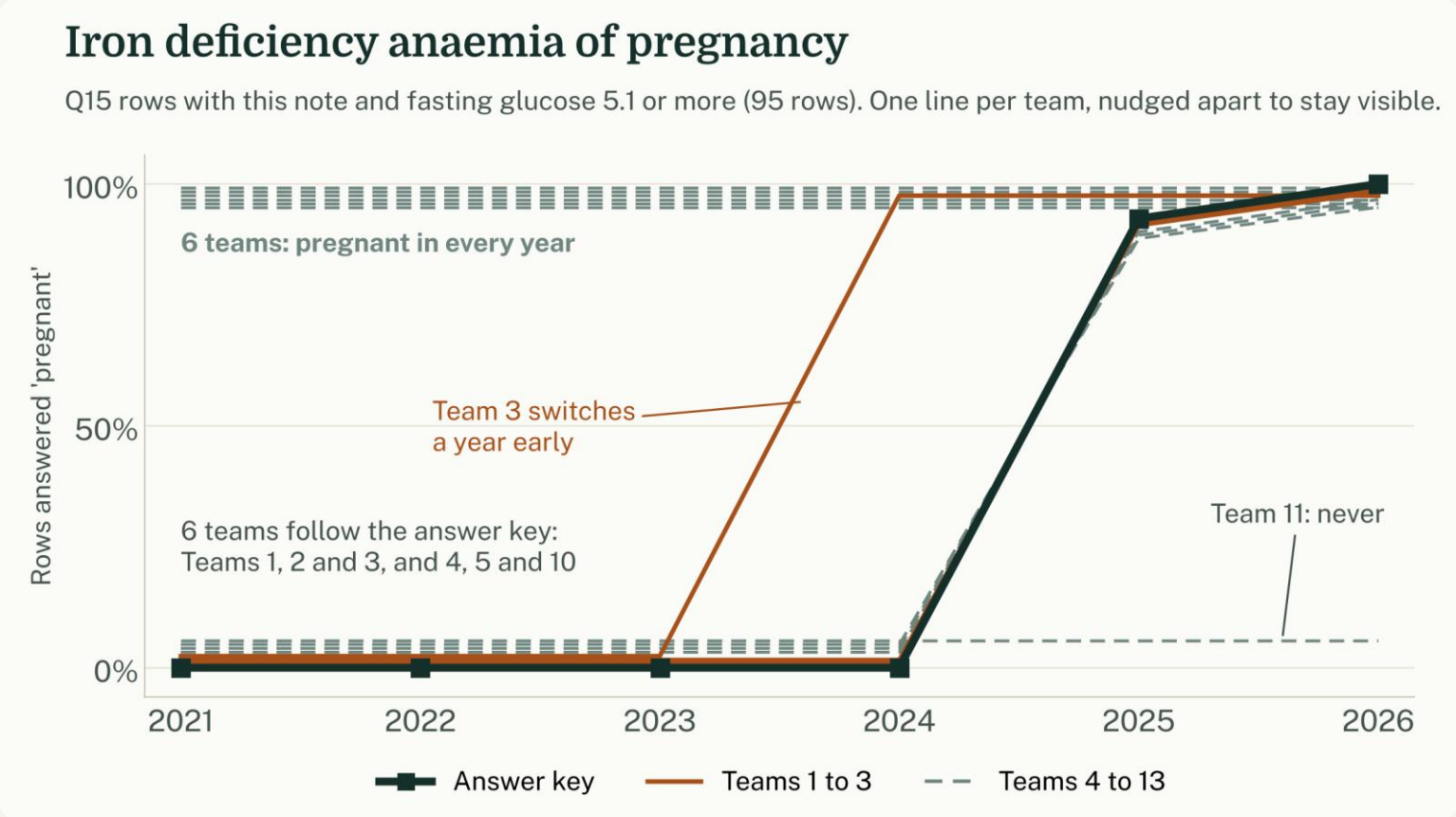
45986006 |Teratogen| has no subtypes, and no substance is defined by the teratogen disposition. Every team built its own list.

Still, **pregnancy** cost more: about 60% of Teams 4 to 13's errors.

The lists split: four teams **added statins**; four **missed antithyroid drugs**.

Q15 · PREGNANT WITH HIGH FASTING GLUCOSE

Right by the rule, wrong in the clinic



Four approaches to one concept.

Our key used **<< Finding related to pregnancy**. Until 2025 this concept had no pregnancy parent, so the key missed it.

Six teams said pregnant every year: clinically right, scored wrong.

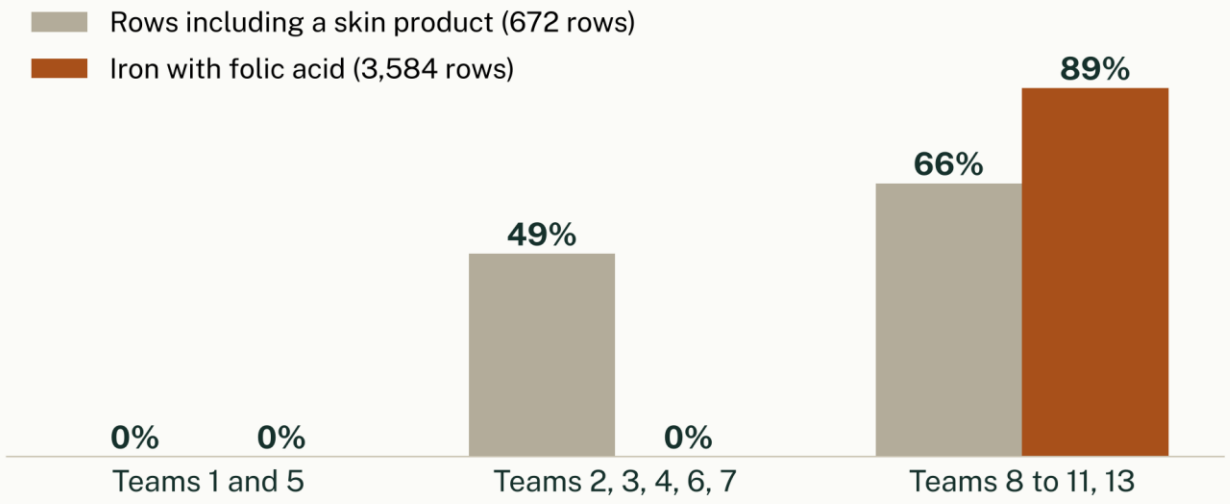
199248002 |Iron deficiency anemia of pregnancy (disorder)|

Q18 · MINERAL SUPPLEMENT

Asked for supplements, scored on ingredients

Which mineral products were missed

Q18: share of answer-key positives each group missed



Team 12 (not shown) missed 80% of other iron products

WHAT TEAMS DID

The key counted zinc oxide nappy cream. **Five teams** dropped about half of the rows with a skin product.

Another five missed **90%** of iron with folic acid products. Two teams got everything.

WHY IT WAS HARD

The question was **silent on specific dose forms**.

373460003 |Mineral supplement| has subtypes, but **none** are used as ingredients of any product.

Participants had to research their own list. ATC A12 is a reasonable start.

|Is modification of| links some salts to their element but not others (zinc oxide, ferrous sulfate), so **query products, not substances**.

Q14 · UPPER LIMB WOUND, NOT FINGERS OR THUMB

Words in the name are not the model



Digital vessel injuries and finger metacarpal fractures are **in**.

Thumb and finger amputations are **out**.

Six teams were perfect; the rest each tripped on a different group.

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FOR YOUR NEXT LONGITUDINAL ANALYSIS

1 Pick an edition strategy. Each one fails differently.

Encounter edition

Keeps codes retired since, as they were meant then.

Inherits that edition's gaps: anaemia of pregnancy was not pregnancy until 2025.

Latest edition

Keeps today's corrected modelling.

Loses retired codes. Q5 falls from 3,819 positives to 748, even with HISTORY-MAX.

Every edition

Keeps both of the above.

Inherits what was later moved out for good reason, such as Thyrotoxicosis from Hyperthyroidism.

Whichever you choose, write it down and test it on the codes that changed.

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2 **Expand and read your value sets.** A descendants query inherits every editorial decision under the parent.

3 **Query attributes, not words.** Finding site, causative agent, ingredient and dose form are usually more robust than a string match.

4 **Write the definition before the ECL.** "Pregnant" and "mineral supplement" are decisions, not queries.

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Thank you for playing

1ST · WINNER

Womp Ratter

Joss Markham and Dougal Fleming

0.9922 macro F1 · top of the private leaderboard

dedalus.com

2ND

Kenny-Bytes

Ken Lee

0.9914 · top of the public leaderboard

makoha.ai

3RD

Marcus Baw

0.9898 macro F1

GitHub: pacharanero
YouTube: Everything Digital Health

Pick an edition strategy, and know how it fails.

Full report, with every question, chart and caveat: kaggle.com/competitions/data-golf-2026